1. **Perform a simple calculation (1. Perform a simple calculation)**
   - The problem requires using the given formula $F = 400\,000y - 3\,200\,000$, setting $F = 0$ to solve for the number of years $y$, which is a simple linear equation-solving process requiring basic algebraic operations (transposition, division) (p. 1).
2. **Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms (5. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms)**
   - Students must understand and apply the specific mathematical formula proposed by the mayor, treating it as a formal system or rule for estimating financial gain. Solving the equation ($400\,000y - 3\,200\,000 = 0$) is an application of a familiar algorithm (p. 1).
3. **Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations (7. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations)**
   - This involves manipulating large numbers (400,000, 3,200,000) and the algebraic equation ($F = ...$) (p. 1).
4. **Make generalisations based on the results of applying mathematical procedures to find solutions (14. Make generalisations based on the results of applying mathematical procedures to find solutions)**
   - After calculating the specific year $y$ (e.g., 8 years) that covers the cost, students need to generalise the "minimum" operational years required and select the correct answer from the options provided, which requires a generalised understanding of the result's meaning (p. 1).
5. **Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation (12. Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation)**
   - Although the mathematical steps to solve this problem are relatively simple, the entire process involves understanding the problem context, identifying relevant information, applying the formula, solving the equation, and finally interpreting the result, which can be seen as a multi-step procedure (p. 1).
6. **Articulate a solution, showing and/or summarising and presenting intermediate mathematical results (8. Articulate a solution, showing and/or summarizing and presenting intermediate mathematical results)**
   - To reach the final answer, the calculation process may need to be written down or performed mentally (e.g., ($y = 3\,200\,000/400\,000$)), which demonstrates the ability to articulate a solution (p. 1).

1. **进行简单计算 (1. Perform a simple calculation)**
   - 该问题要求使用给定的公式 $F = 400\,000y - 3\,200\,000$ (p. 1)，通过设置 $F = 0$ 来求解年份 $y$，这是一个简单的一元一次方程求解过程，需要基础的代数运算（移项、除法）。
2. **理解和利用基于定义、规则和形式系统的结构，包括使用熟悉的算法 (5. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms)**
   - 学生必须理解并应用市长提出的特定数学公式，将其视为一个用于估计财务收益的形式系统或规则 (p. 1)。解方程 ($400\,000y - 3\,200\,000 = 0$) 是一个熟悉的算法应用。
3. **操作数字、图形和统计数据及信息、代数表达式和方程式、以及几何表示 (7. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations)**
   - 这涉及操作大数字（400,000、3,200,000）和代数方程式 ($F = \ldots$) (p. 1)。
4. **根据应用数学程序的结果进行概括 (14. Make generalisations based on the results of applying mathematical procedures to find solutions)**
   - 在计算出特定年份 $y$（例如 8 年）使成本得以覆盖后，学生需要概括出所需的“最小”操作年限，并从给定的选项中选择正确答案，这需要对结果的含义进行概括性理解 (p. 1)。
5. **使用多步骤程序得出数学解、结论或概括 (12. Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation)**
   - 虽然解决这个问题的数学步骤相对简单，但整个过程涉及理解问题情境、识别相关信息、构建或应用公式、求解方程，最后解释结果，可被视为一个多步骤的过程。
6. **阐明解决方案，展示和/或总结并呈现中间数学结果 (8. Articulate a solution, showing and/或 summarizing and presenting intermediate mathematical results)**
   - 为了得出最终答案，可能需要写出或在脑海中完成计算过程（例如 ($y = 3\,200\,000/400\,000$)），这体现了阐明解决方案的能力。